

APSC 2113 - Engineering Analysis I
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LQ1 - Fall 2026

Questions

Q1. Consider the differential equation

$$\frac{dy}{dx} + x\frac{d^2y}{dx^2} + y = 0.$$

What are the **order** and **linearity** of this differential equation?

- A. First order and linear
- B. First order and nonlinear
- C. **Second order and linear (Correct Solution)**
- D. Second order and nonlinear

Explanation:

The highest derivative present is $\frac{d^2y}{dx^2}$, making the equation second order.

All dependent terms (y , $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$) appear to the first power with coefficients independent of y , making it linear.

Q2. Find the general solution of the differential equation

$$\frac{dy}{dx} = 4y.$$

- A. $y = 4e^x + C$
- B. $y = Ce^{4x}$ **(Correct Solution)**
- C. $y = Ce^x + 4$
- D. $y = 4Ce^x$

Explanation:

Separating variables gives $\frac{1}{y}dy = 4dx$, which integrates to $\ln|y| = 4x + C_1$.

Exponentiating both sides yields $y = e^{4x+C_1} = Ce^{4x}$.

Q3. Evaluate

$$\int (3e^{2x} - 4e^{-x}) dx.$$

- A. $\frac{3}{2}e^{2x} + 4e^{-x} + C$ **(Correct Solution)**
- B. $6e^{2x} - 4e^{-x} + C$
- C. $\frac{3}{2}e^{2x} - 4e^{-x} + C$
- D. $3e^{2x} + 4e^{-x} + C$

Explanation:

Integrating term-by-term using $\int e^{kx} dx = \frac{1}{k}e^{kx}$ gives $\frac{3}{2}e^{2x} - 4\left(\frac{e^{-x}}{-1}\right) + C$.

Simplifying the second negative sign yields $\frac{3}{2}e^{2x} + 4e^{-x} + C$.

Q4. The function $y = Ce^{-3x}$ satisfies the differential equation $\frac{dy}{dx} = -3y$.

A. **True** (Correct Solution)

B. False

Explanation:

Taking the derivative yields $\frac{dy}{dx} = -3Ce^{-3x}$.

Substituting $y = Ce^{-3x}$ gives $\frac{dy}{dx} = -3y$, which verifies the statement as True.

Q5. If $N(t) = N_0e^t$, where n is a positive integer, what is $\frac{d^n N(t)}{dt^n}$?

A. N_0e^{nt}

B. N_0e^t (Correct Solution)

C. nN_0e^t

D. $(n-1)N_0e^t$

Explanation:

The derivative of e^t with respect to t is always e^t for any order n .

Thus, differentiating N_0e^t repeatedly n times leaves the expression unchanged as N_0e^t .
